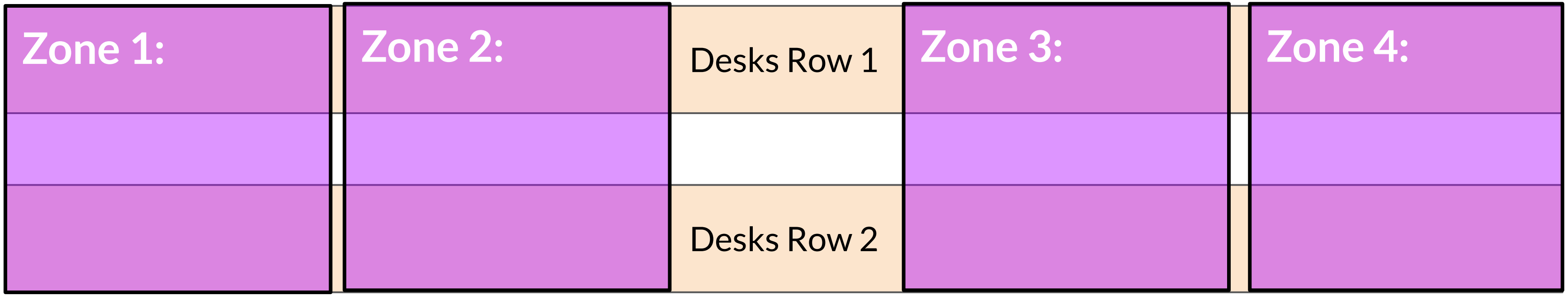
CSE 151B: Deep Learning

Instructor: Annapurna Vadaparty

TA: Ben Bao

Tutors: Alexander Radulescu, Trevor Duong

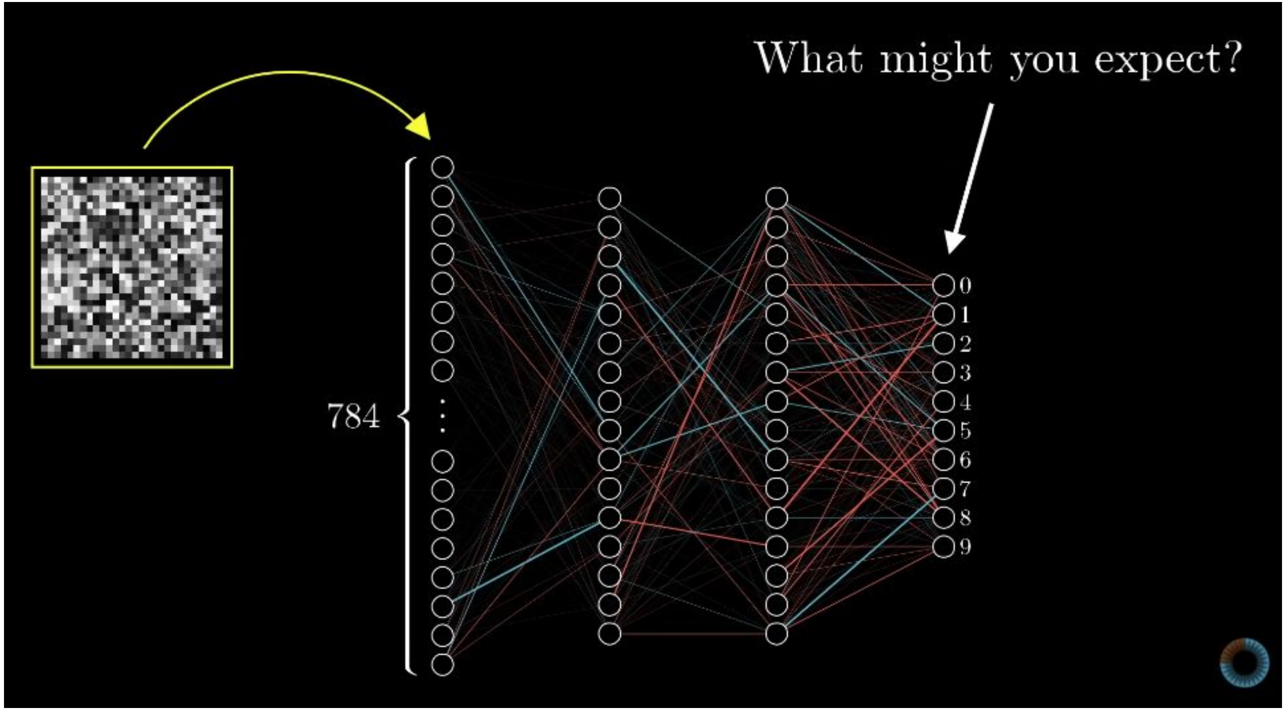
Seating: We'll start assigning next week.



Start talking to the people in your zone about the following (after you know their names!):

At the end of the "Gradient descent, how neural networks learn" 3b1b video, he points out that if you train a neural network on random noise, it will give a confidently wrong answer. Why does it do this instead of unconfident, diffuse activations?

What happens to a NN when you input random noise?



3b1b: Gradient descent, how neural networks learn

The importance of data sourcing

"From its point of view, the entire universe consists of nothing but clearly defined, unmoving digits centered in a tiny grid.

And its cost function just never gave it any incentive to be anything but utterly confident in its decisions."

Questions from yesterday?

- Pre-lecture homeworks? Due daily
- Assignment 0? Due tomorrow
- Pre-course survey/textbook access?
- Signing up for Oral Exams? Due tonight

- Practice exam through the CBTF? Required by Jul 2

Release: start of today	Opens
Early 1: end of today	100%
Due: Fri, Jul 3, end of day	100%
Late 1: Sat, Jul 4, end of day	99%

Why do we need neural networks?

There are lots of ways to handle linear boundaries between classes.

AND

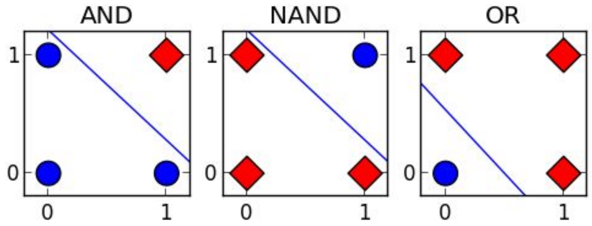
Input 1	Input 2	Output
0	0	●
0	1	●
1	1	◆
1	0	●

OR

Input 1	Input 2	Output
0	0	●
0	1	◆
1	1	◆
1	0	◆

NAND

Input 1	Input 2	Output
0	0	◆
0	1	◆
1	1	●
1	0	◆



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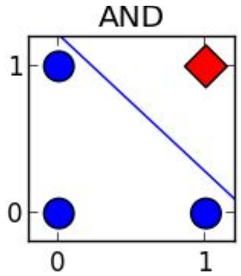
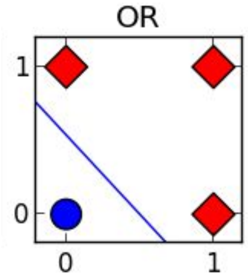
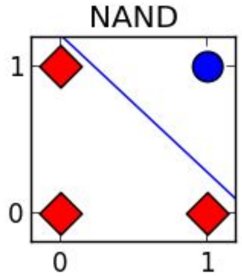






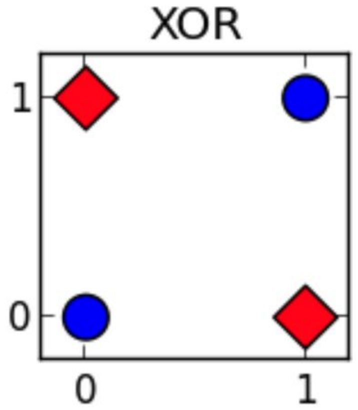
Why do we need neural networks?

- What happens when a line isn't able to separate your classes?
- Let's say you wanted to classify something as simple as XOR:

Input 1	Input 2	Output
0	0	0
0	1	1
1	1	0
1	0	1

XOR

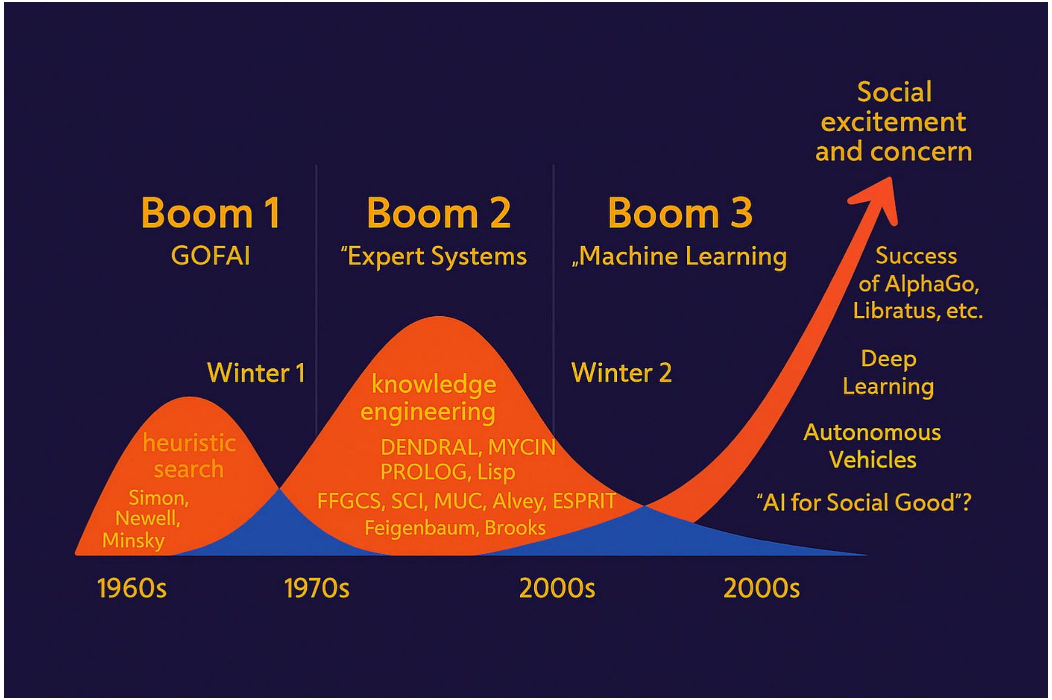
Input 1	Input 2	Output
0	0	●
0	1	◆
1	1	●
1	0	◆



This was interesting to people working on AI…

- 1969: Marvin Minsky and Seymour Papert pointed out a major weakness of early neural networks
    - Single-layer networks are just linear classifiers–one weight matrix times an input matrix
- Single-layer perceptron can't solve the XOR problem
- Interest and funding for neural network research dropped sharply.
- This slowdown became known as the first "AI Winter."

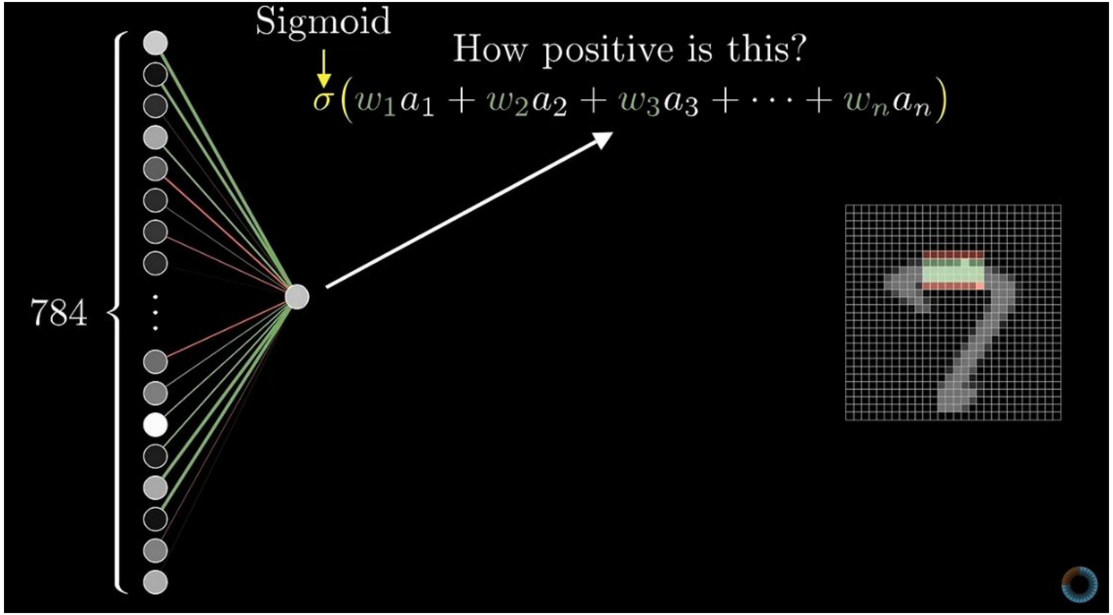
AI Winters



https://pub.towardsai.net/what-is-ai-winter-understanding-the-causes-behind-the-decline-in-ai-research-440ec81b3866

Neural networks let you handle really any decision boundary.

Structure of a Neural Network



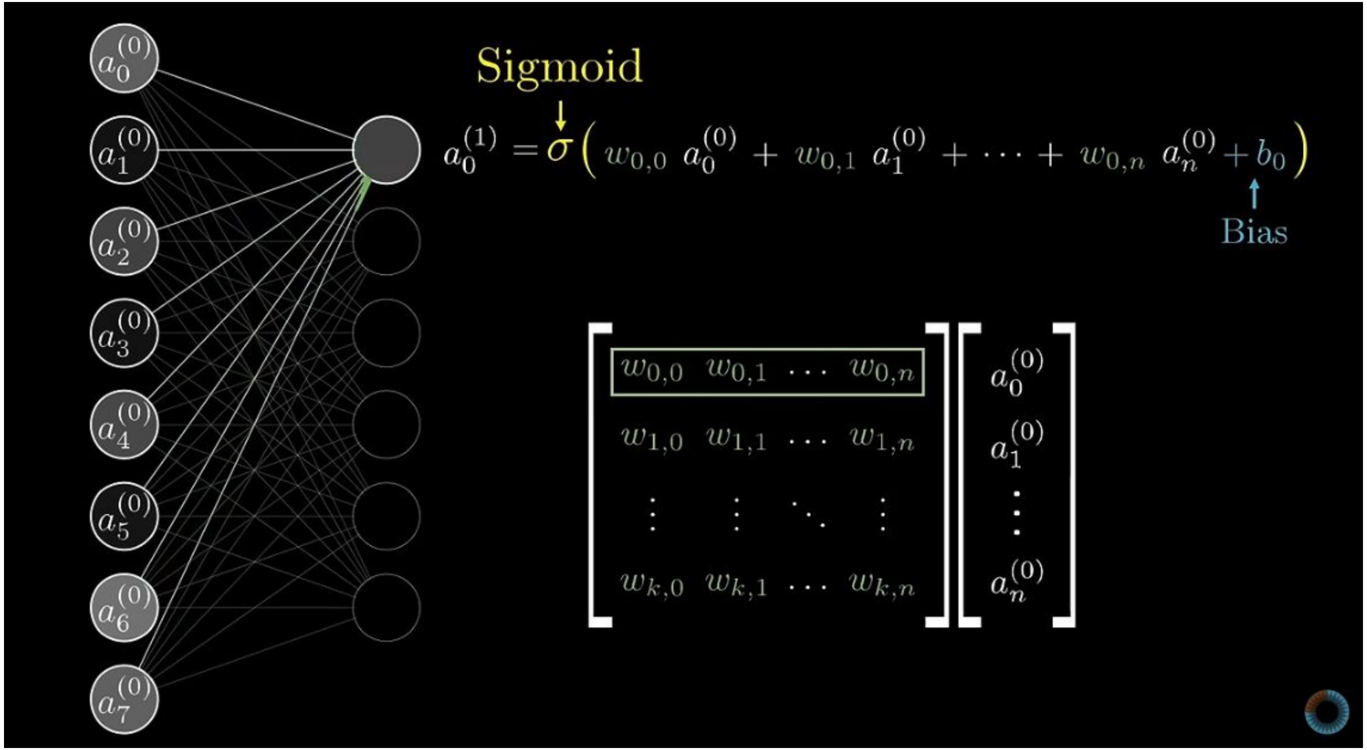
Note: We see sigmoids a lot in DL! Where is it used?

Nowadays, it is usually just a “squishification” function (keeps values between 0 and 1), but not activation.

In the video, he uses it to function as both squishification and activation.

3b1b: But what is a neural network?

It’s helpful to draw out matrices like this!



3b1b: But what is a neural network?

Break!
See you in 5 minutes.

Tomorrow, we’ll learn how to cleverly update neural networks.

To prepare for that, we’ll want to represent each step and dependency very clearly.

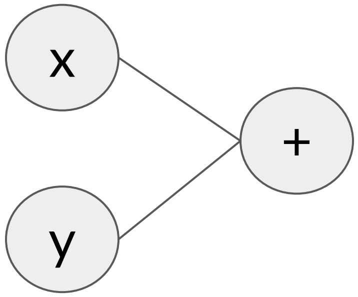
A way to represent each step: computation graphs

What is a computation graph?

- Nodes: Variables (inputs, weights, biases) or mathematical operations (+, -, x, /, sin, cos, etc.)

- Edges: Flow of data (tensors/matrices) as they move from one operation to the next.

$f(x, y) = x + y$



A way to represent each step: computation graphs

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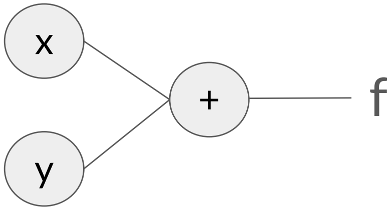
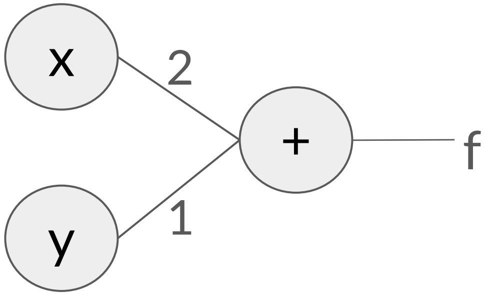


$x = 2, y = 1$

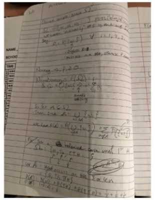
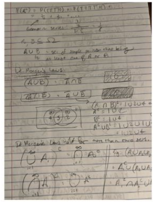


Now try with: $f(x, y, z) = (x + y) * z$

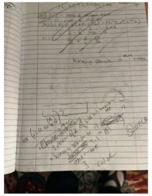
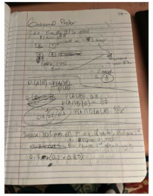
- Draw it in your notebook!
- When you've finished, compare with the person next to you. What do your solutions look like?

Roughbooks are...rough!

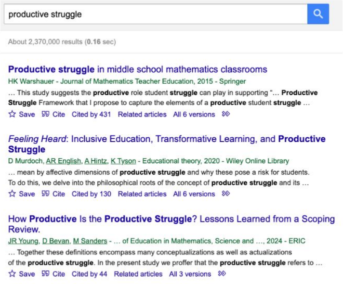
They make no promises about handwriting...

Or correctness...

Their promise is to help you learn.



productive struggle

About 2,370,000 results (**0.16** sec)

Productive struggle in middle school mathematics classrooms
HK Warshauer - Journal of Mathematics Teacher Education, 2015 - Springer
… This study suggests the **productive** role student **struggle** can play in supporting "… **Productive Struggle** Framework that I propose to capture the elements of a **productive** student **struggle** …
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Feeling Heard: Inclusive Education, Transformative Learning, and **Productive Struggle**
D Murdoch, AR English, A Hintz, K Tyson - Educational theory, 2020 - Wiley Online Library
… mean by affective dimensions of **productive struggle** and why these pose a risk for students. To do this, we delve into the philosophical roots of the concept of **productive struggle** and its …
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How **Productive** Is the **Productive Struggle**? Lessons Learned from a Scoping Review.
JR Young, D Bevan, M Sanders - … of Education in Mathematics, Science and …, 2024 - ERIC
… Together these definitions encompass many conceptualizations as well as actualizations of the **productive struggle**. In the present study we proffer that the **productive struggle** refers to …
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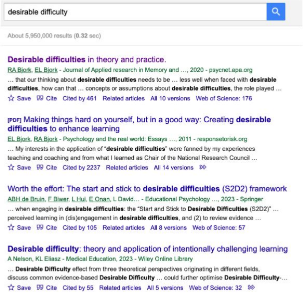

desirable difficulty

About 5,950,000 results (**0.32** sec)

Desirable difficulties in theory and practice.
RA Bjork, EL Bjork - Journal of Applied research in Memory and …, 2020 - psycnet.apa.org
… that our thinking about **desirable difficulties** needs to be … less well when faced with **desirable difficulties**, how can that … concepts or assumptions about **desirable difficulties**, the role played …
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[PDF] Making things hard on yourself, but in a good way: Creating **desirable difficulties** to enhance learning
EL Bjork, RA Bjork - Psychology and the real world: Essays …, 2011 - responsetorisk.org
… My interests in the application of "**desirable difficulties**" were fanned by my experiences teaching and coaching and from what I learned as Chair of the National Research Council …
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Worth the effort: The start and stick to **desirable difficulties** (S2D2) framework
ABH de Bruin, F Biwer, L Hui, E Onan, L David… - Educational Psychology …, 2023 - Springer
… when engaging in **desirable difficulties**: the "Start and Stick to **Desirable Difficulties** (S2D2)" … perceived learning in (dis)engagement in **desirable difficulties**, and (2) to review evidence …
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Desirable difficulty: theory and application of intentionally challenging learning
A Nelson, KL Eliasz - Medical Education, 2023 - Wiley Online Library
… **Desirable Difficulty** effect from three theoretical perspectives originating in different fields, discuss common evidence-based **Desirable Difficulty** … could further optimise **Desirable Difficulty**-…
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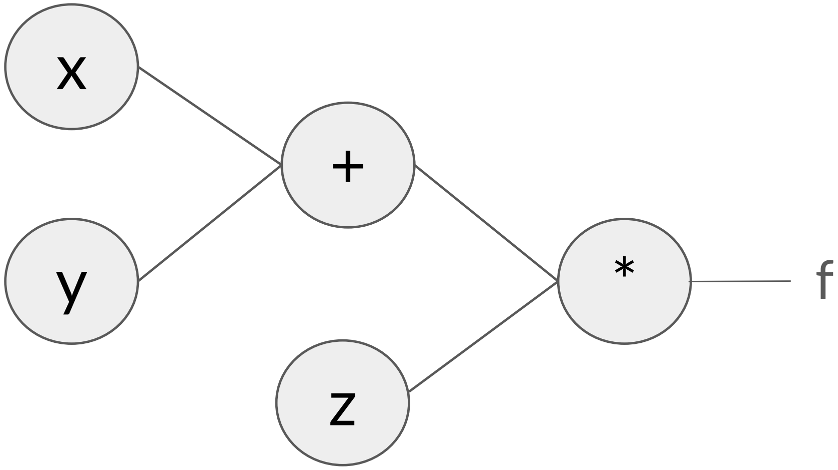
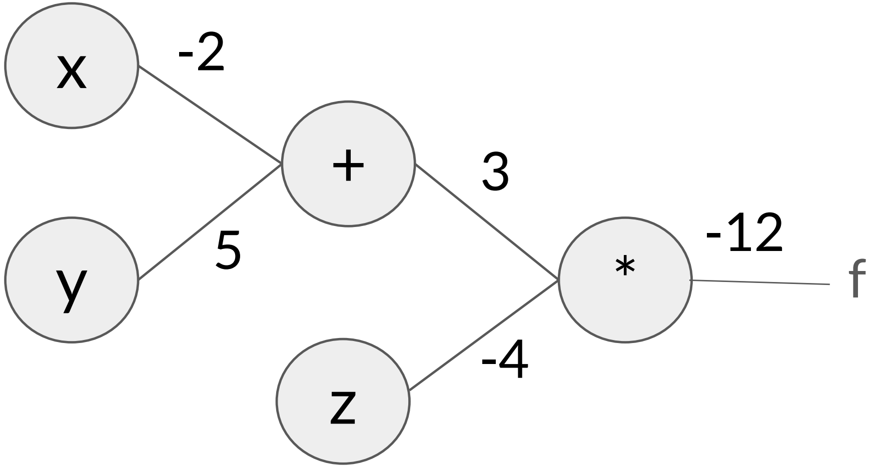
Given inputs x = -2, y = 5, z =-4:



Add them in your notebooks, and someone add it to the board!

Given inputs x = -2, y = 5, z =-4:



How about in a neural network with weights and activation?

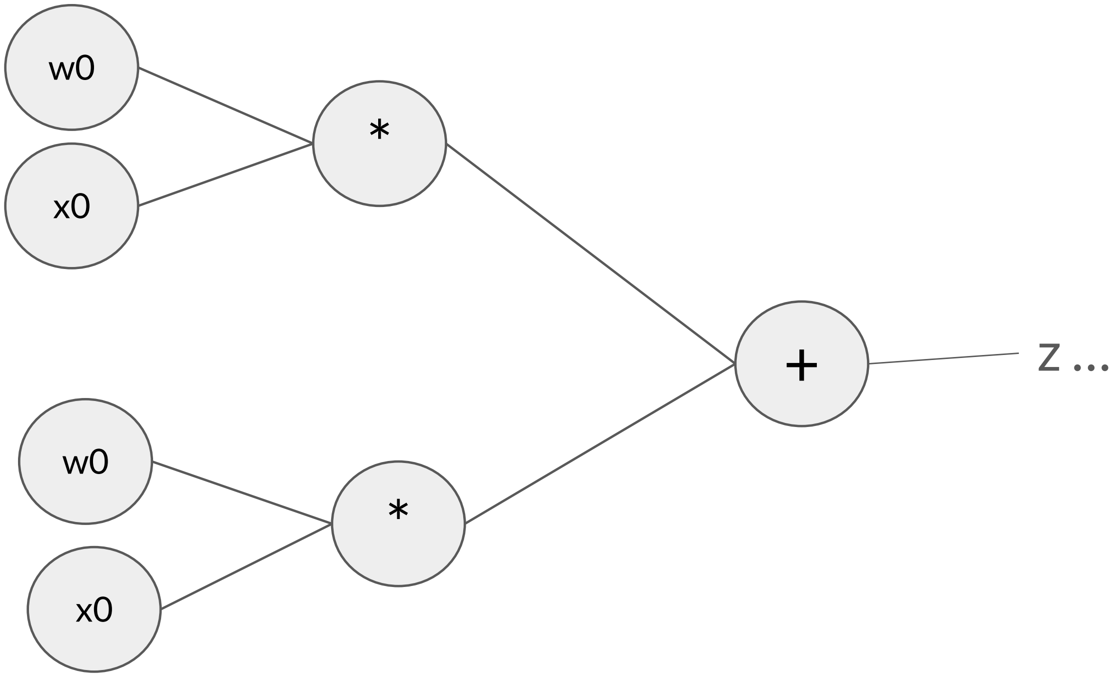
First: $z = w_0x_0 + w_1x_1 + b$

Then: $a = \sigma(z) = \frac{1}{1 + e^{-z}}$

- Draw it in your notebook! (Let’s see how far we get in 5 minutes)
- When you’ve finished, compare with the person next to you. What do your solutions look like?

Let’s draw it on the board…

Beginning of first step...



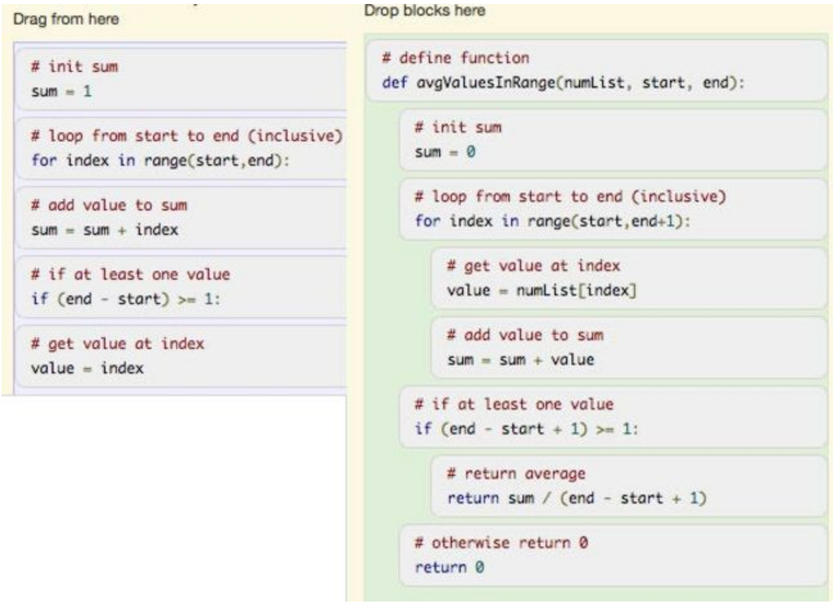
Next steps to finish the computation graph?

You’re welcome to think about this before class tomorrow…we’ll spend some time getting that computation graph ready before we do backpropagation with it!

Assignments are Autograded (unlike what we just did)

- Gives you immediate feedback to learn from
- Avoids grading time for teaching staff
- Free response converted to:
  - Parsons problems
  - Multiple choice
  - Fill in the blank

Parsons problems



But you should work on them yourself first!

- PDFs of open-ended versions of problems posted
- You can think of the PrairieLearn formulation (Parsons, MC) as scaffolding to help you out if you're not sure how to start
  - In PL make sure you scroll sideways for the Parsons problems – not everything shows in each block without scrolling
- Try them yourself first like what we just did!

Operationalizing “work on them yourself first”

Doing problems yourself on paper is probably less frustrating anyway!

- You’ll feel like you know what’s going on rather than trying to figure out what each Parson’s block is trying to do (like trying to debug code you’re not familiar with).
- But don’t feel bad if you need to look! This probably just means that once you’re done, it’s a good idea to go find other practice problems like it.

Sample Oral Assessment Questions for A0: Understand

```
# Initialize an example tensor
x = torch.Tensor([
                  [[1, 2], [3, 4]],
                  [[5, 6], [7, 8]],
                  [[9, 10], [11, 12]]
                 ])
x
```

```
tensor([[[ 1.,  2.],
         [ 3.,  4.]],

        [[ 5.,  6.],
         [ 7.,  8.]],

        [[ 9., 10.],
         [11., 12.]]])
```

```
x.shape
```

```
torch.Size([3, 2, 2])
```

```
# Access the 0th element, which is the first row
x[0] # Equivalent to x[0, :]
```

```
tensor([[1., 2.],
        [3., 4.]])
```

(Just for this week, these will be questions with just a numerical answer. Otherwise this would turn into testing you on syntax and/or doing matrix operations in your head.)

- How many dimensions are in this tensor?
- What is the shape of the [3, 4] element?
- What is the shape of x[0]?

Sample Oral Assessment Questions for A0: Experiment

```
# Initialize an example tensor
x = torch.Tensor([
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                  [[5, 6], [7, 8]],
                  [[9, 10], [11, 12]]
                 ])
x
```

```
tensor([[[ 1.,  2.],
         [ 3.,  4.]],

        [[ 5.,  6.],
         [ 7.,  8.]],

        [[ 9., 10.],
         [11., 12.]]])
```

```
x.shape
```

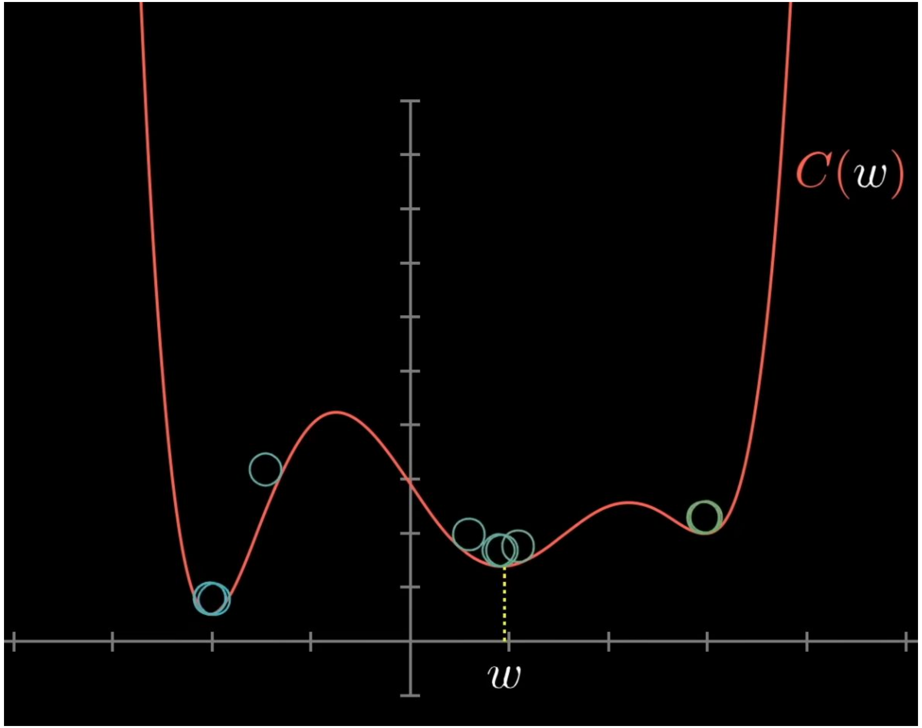
```
torch.Size([3, 2, 2])
```

```
# Access the 0th element, which is the first row
x[0] # Equivalent to x[0, :]
```

```
tensor([[1., 2.],
        [3., 4.]])
```

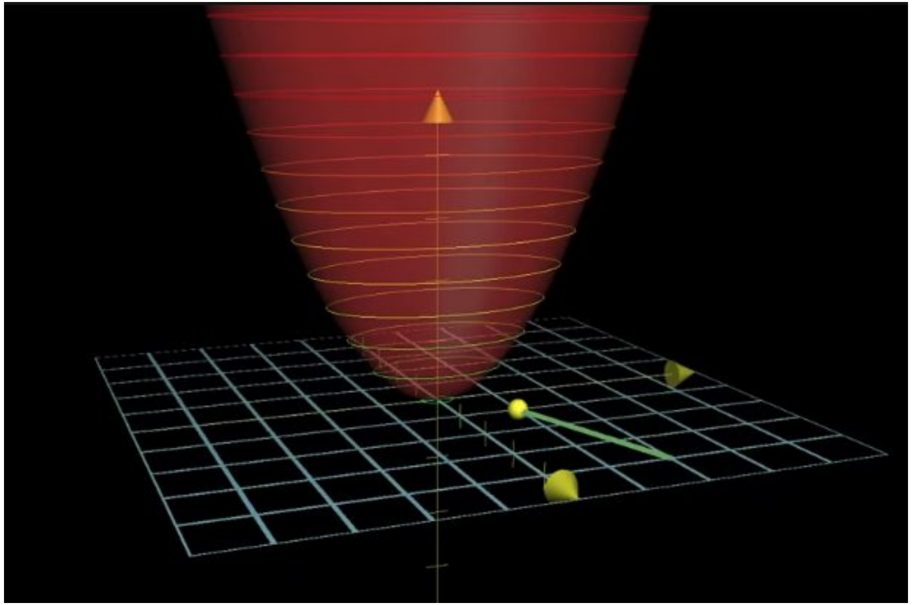
- What would happen if I indexed at x[1] instead?
- What would happen if I inserted a single number in a list, e.g. changing [1,2] to [1, 2, 7] and didn't change anything else?

Gradient Descent



3b1b: Gradient descent, how neural networks learn

What does a gradient mean?

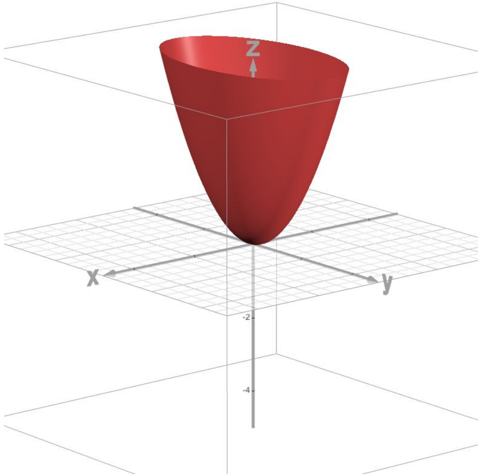


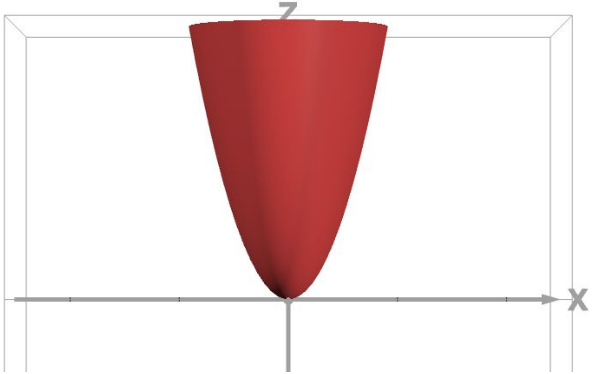
3b1b: Gradient descent, how neural networks learn

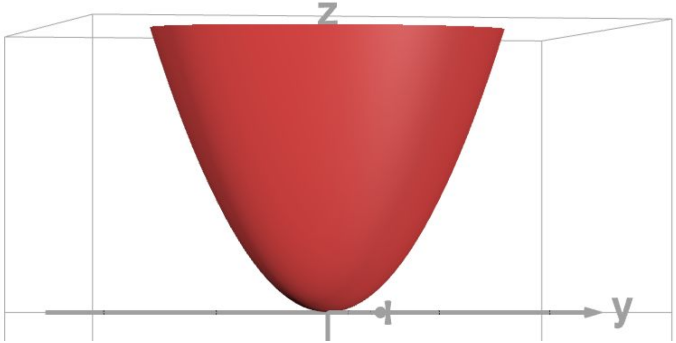
Let’s look at this manifold

- https://www.desmos.com/3d

$$z = \frac{3}{2}x^2 + \frac{1}{2}y^2$$







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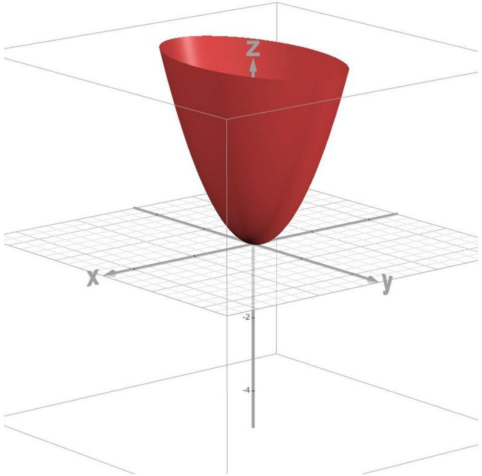
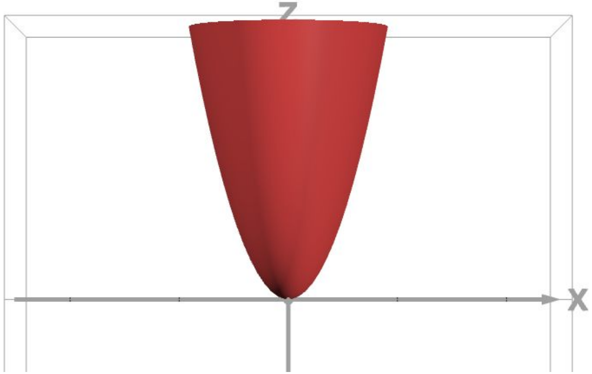
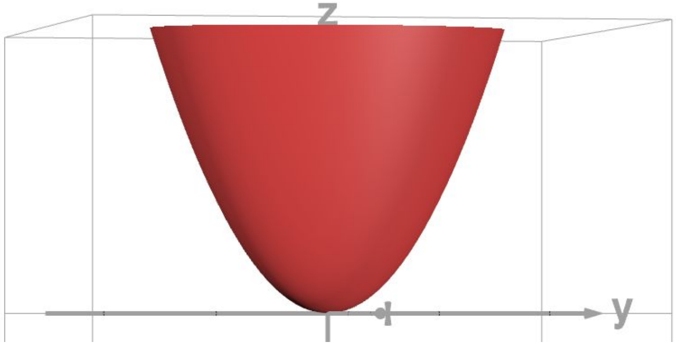






You only have to move a little bit in the x direction to get a large change in the z direction

You have to move further in the y direction to get that same change in the z direction

End of where we reached in class

How would you communicate this in math?

- “You only have to move a little bit in the x direction to get a large change in the z direction”
  - Moving 1 unit in the x direction gets you 3 unit up in the z direction
- “You have to move further in the y direction to get that same change in the z direction”
  - Moving 1 unit in the y direction gets you 1 unit up in the z direction

Gradient

How did we communicate this in 2D equations in single-variable calculus?

- A line $y=x^2$ has a derivative of 2x. We just have the one expression, 2x.

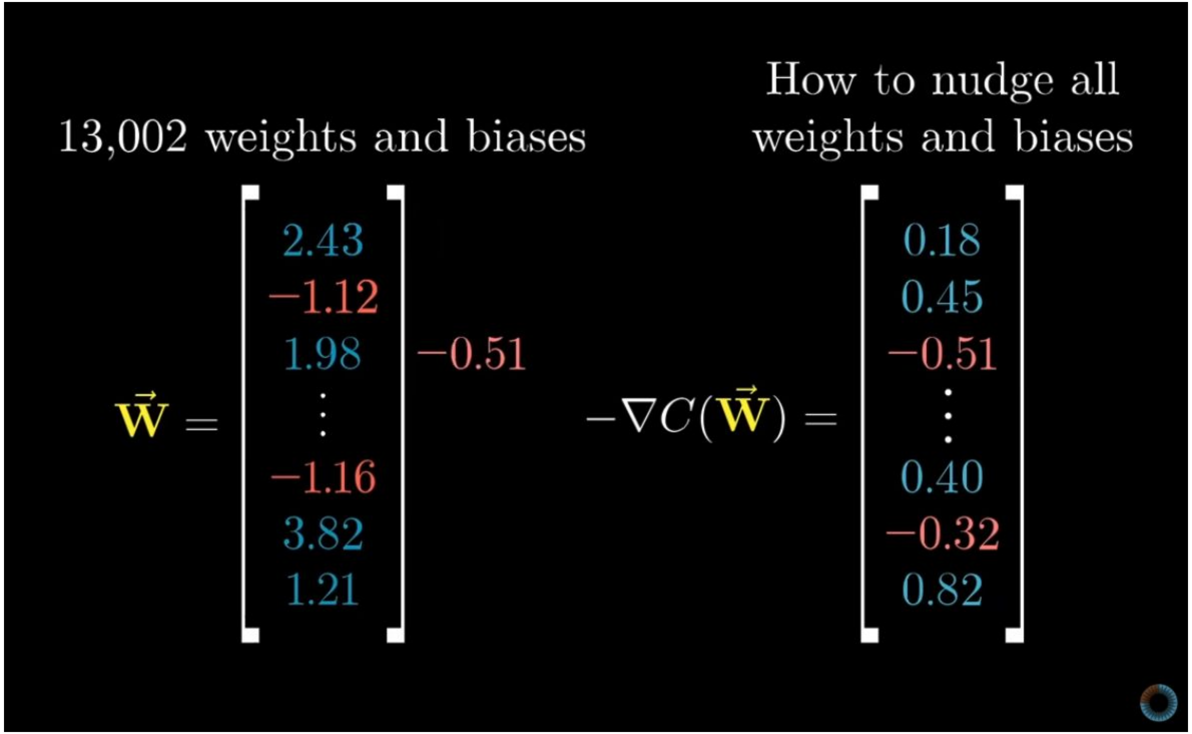
What do we do when we have two dimensions to account for?

- Gradient! Two values we store in a vector: Derivative w.r.t. x is 3x, w.r.t y is y. This communicates what we wanted: our vector below tells us exactly how much the gradient changes with each a given change in x and y.

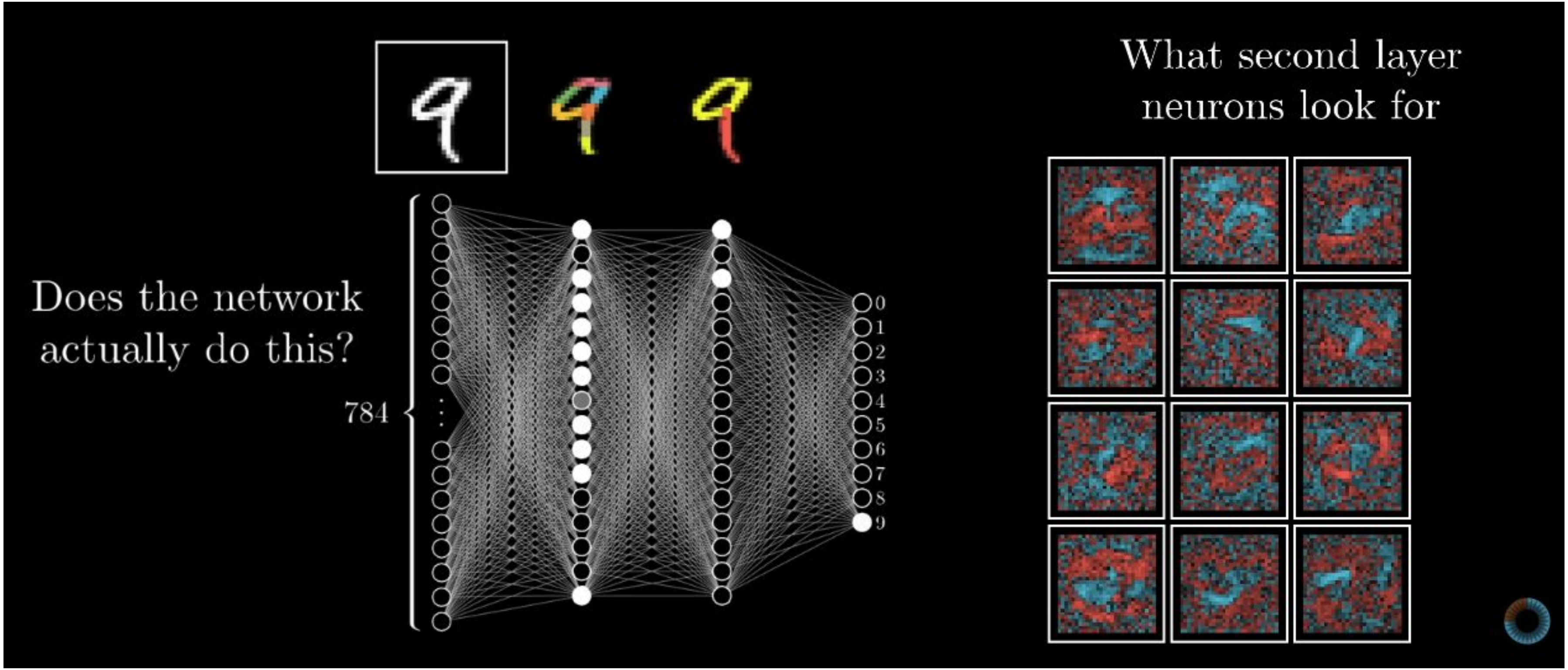
$$\nabla f(x, y) = (3x, y)$$

- A vector is just a way of storing values in multiple dimensions. All the math is the same as what you first learned in calculus; vectors are a helpful way to keep track.

Gradient descent in a neural network



What is the network learning?



3b1b: Gradient descent, how neural networks learn

Answers to open questions from class

- What happens in computation graphs for binary operators (operators that take two inputs) whose input order matters? (e.g. subtraction, division)
    - Some references (e.g. Karpathy's micrograd) do add + negate, while others (e.g. the Pytorch docs) keep subtraction as a primitive and expect inputs in a certain order to resolve this ambiguity.
    - For teaching purposes, there doesn't seem to be a ground truth either–some resources include negatives (e.g. these slides) whereas others expect students not to use these operators
    - The takeaway here is that it depends on the notation you and the reader agree on–as long as you're consistent, either works.
- How complex of an operator can we represent in a node (e.g. can the entire sigmoid function be represented in one node)?
    - Composites like sigmoid can be represented in one node, but generally we expand it into simpler components because having "easy" derivatives makes it easier to follow, which is useful for debugging purposes (which is a major use of computational graphs in practice).